

CURRICULUM VITAE

André NICOLET

Full Professor (Professeur des Universités, Classe Exceptionnelle)
Institut Fresnel (CNRS UMR 7249) – Aix-Marseille Université

1 CAREER

Professional address

Institut Fresnel, Aix-Marseille Université
Faculté des Sciences de Saint-Jérôme – case 162
Avenue Escadrille Normandie-Niemen
F-13397 Marseille cedex 20, France

Online

Web page: andre-nicolet.github.io
Google Scholar: [profile](#)
E-mail: [andre.nicolet \(at\) fresnel.fr](mailto:andre.nicolet@fresnel.fr)

1.1 Degrees

- Electrical Engineer (*Ingénieur Civil Électricien*, electronics), July 1984, *magna cum laude* (Grande Distinction), University of Liège, Belgium.
- PhD in Applied Sciences, defended on 29 November 1991, *summa cum laude* (la Plus Grande Distinction), University of Liège.
Thesis committee: Professors G. Cantraine (University of Liège, chair), W. Geysen (Katholieke Universiteit Leuven), M. Hogge (University of Liège), W. Legros (University of Liège), D. Lowther (McGill University, Montréal), M. Pavella (University of Liège), P. Pirotte (University of Liège).
- *Habilitation à Diriger des Recherches* (French habilitation to supervise research, 63rd section), defended on 18 January 1996, Université d'Aix-Marseille III.
Committee: Professors A. Genon (University of Liège), C. Glaize (Université de Montpellier II), M. Lajoie-Mazenc (INP Toulouse), A. Nicolas (École Centrale de Lyon, referee), F. Piriou (Université des Sciences et Technologies de Lille, referee), J. C. Sabonnadière (ENSIEG, Grenoble, chair and referee), M. Tholomier (Université d'Aix-Marseille III).

1.2 Current position

Since September 1996: Full Professor (*Professeur des Universités*, 63rd section: Electronics, Optronics and Systems) at Université d'Aix-Marseille III, now Aix-Marseille Université; promoted to the highest grade (*Classe Exceptionnelle*) on 1 September 2019. Member of Institut Fresnel (CNRS UMR 7249) since its creation in 2000.

1.3 Previous positions

- November 1994 – August 1996: Head of the Department of Electrical Engineering and Power Electronics, École Supérieure d'Ingénieurs de Marseille (ESIM);
- April 1992 – October 1994: Senior Research Engineer in the group of Prof. W. Legros, University of Liège, within the Interuniversity Attraction Pole (P.A.I.) *CAD of Electromagnetic Systems*;
- September 1988 – March 1992: Research Engineer in the group of Prof. W. Legros, within the same Interuniversity Attraction Pole;

- July 1987 – August 1988: Research Engineer in the group of Prof. W. Legros, on a contract with the Walloon Region entitled *Determination of magnetic field lines in nonlinear materials*;
- October 1984 – June 1987: Researcher in the Applied Electricity group of Prof. W. Legros, University of Liège, on a PREST (S.P.P.S.) contract on the *Computation of the space-time evolution of magnetic fields*.

1.4 Languages

- French: native language;
- English: fluent (spoken, read and written);
- Dutch: good comprehension;
- German: beginner.

1.5 Award

S.R.B.E. Prize (Royal Belgian Society of Electrical Engineers), 1992, for the doctoral thesis *Modelling of the magnetic field in systems including nonlinear media*.

2 SCIENTIFIC PROJECT

At the beginning of my career, I developed codes for the computation of low-frequency electromagnetic fields for electrical engineering applications. Having used first integral methods, then finite elements, including the coupling of both methods, I acquired a solid background in applied mathematics and numerical analysis. Drawn to high-frequency modelling by the emergence of electromagnetic compatibility problems, I enthusiastically joined Institut Fresnel at its creation in 2000. With F. Zolla and S. Guenneau, I applied the finite element method to microstructured optical fibres, which requires both the full set of Maxwell's equations (a wave propagation phenomenon) and the computation of eigenvalues of operators (a modal problem). This research, carried out with the other members of the CLARTE team, led to the book *Foundations of Photonic Crystal Fibres*, a reference in the field, whose augmented second edition was published in 2012. We studied **twisted fibres** and developed a numerical technique based on a non-orthogonal coordinate system. Building on my background in differential geometry and on intuitions going back to the 1990s, based on the fact that Maxwell's equations are independent of the metric, which only enters the constitutive relations¹, I stated as early as 2003 the principle that any change of coordinates can be encoded as equivalent materials: **transformation optics** was born well before the media coverage of J. Pendry's **invisibility cloak** in 2006. We were thus naturally among the first to publish on these topics, in particular in collaboration with J. Pendry himself. Since then, the finite element method, which was hardly used in optics laboratories when I arrived at Institut Fresnel, has become an essential tool in the laboratory. Perfectly Matched Layers (PMLs), which can be interpreted within transformation optics as complex-valued changes of coordinates, make it possible to handle wave problems in open domains correctly, not only for scattering but also for modal analysis (leaky modes of waveguides and quasi-modes of structures such as diffraction gratings). These numerical modal analyses naturally lead us to work with Hilbert spaces, the operators defined on them and their spectra: we are not far from the formalism of quantum mechanics². To explore this connection, I co-supervised with Prof. T. Durt the PhD thesis of Vincent Debierre, *The photon wave function, in principle and in practice* (École Centrale de Marseille, defended in September 2015), which studies the boundary between classical and quantum electrodynamics by critically discussing the link between the classical electromagnetic field and a possible **photon wave function**.

One application of this approach is the possibility of modelling quantum electrodynamics devices with classical models. With G. Demésy, we studied the electromagnetic modelling of the **QED cavity of the Haroche** (Nobel Prize in Physics 2012)–Raimond–Brune team at LKB, to show that the photon lifetime is bounded by the characteristics of a quasi-mode that can be computed with finite elements and PMLs. This computation requires high-order finite elements, and the study served as a test case for the PhD thesis of Nicolas Marsic (supervised by Prof. Geuzaine, University of Liège, defended in January 2016).

This work is a direct continuation of the **modal analysis of open structures** developed during the PhD thesis of B. Vial (defended in 2013). This tool is under active development from the theoretical (understanding the discretisation of the continuous spectrum), numerical (extension to dispersive materials leading to nonlinear eigenvalue problems) and applied (efficient computation of LDOS) points of view.

This activity led in 2016 to an ANR project, of which I was the local principal investigator, on **quasi-modal analysis**, with LP2N, Institut d'Optique d'Aquitaine (Ph. Lalanne) and Institut d'Optique, Palaiseau (C. Sauvan). In 2018, we removed the main obstacle to using this approach in highly dispersive media, thanks to a universal formulation of the modal expansion in arbitrary dispersive media (published in *Optics Letters*, December 2018). The ANR project ended in 2020 with results assessed as very positive, and the topic is still very active within the ATHENA team.

¹We are not far from the concepts of general relativity.

²In Wigner's classification, Maxwell's equations indeed appear as the Schrödinger equation of massless (hence necessarily relativistic) spin-1 particles.

Creation of a research team (ATHENA)

In March 2015, I **founded and headed the ATHENA team** at Institut Fresnel for a four-year term. Besides myself, its members are Prof. F. Zolla, Prof. G. Renversez and Dr. G. Demésy. The central theme of ATHENA is the study of “structurally complex” electromagnetic problems by theoretical and numerical methods. By structurally complex, we mean problems whose geometry and material properties are essentially arbitrary.

From the geometrical point of view, the objects studied may have arbitrary shapes (not only canonical shapes such as spheres, cylinders or cubes), may be reduced to two dimensions (very thin layers, membranes) or one dimension (wires), and may be periodic and/or unbounded in one or several directions. From the point of view of material properties, we consider anisotropic and inhomogeneous media. These properties may be intrinsic to a natural medium or result from a homogenisation process. Transformation optics (to handle special geometries such as twisted objects, or to introduce PMLs, a fundamental tool for unbounded problems, through complex-valued changes of coordinates) also leads to such media. We also consider nonlinear media whose properties depend on the local electromagnetic field. In all these cases, the best-suited numerical methods are “domain” methods such as finite elements (edge elements, in their modern form, for vector fields), finite differences or discontinuous Galerkin methods. Until the finite element method is generalised to the nonlinear case in the time domain, we use the latest developments of the FDTD method for this case.

Our approach is part of the free and open-source software movement, which serves as the basis for developing our own codes and models. This is made possible by our many fruitful national and international collaborations. **This research has been disseminated through a free software library of models: ONELAB Photonics (Open Source Finite Element Software for Photonics Applications, <https://onelab.info/photonics/>).** Each model is fully editable and can serve as a basis for users’ own models. It is part of the ONELAB project of Prof. C. Geuzaine (University of Liège) and uses the SLEPc library of Jose E. Roman (Universitat Politècnica de València) for eigenvalue computations (including nonlinear ones for dispersive media). The development of our models has led to corrections and improvements in ONELAB and SLEPc.

One of the main research lines is the **modal analysis of open problems**. Combining finite elements and PMLs, one can determine the “leaky” resonances (with complex frequencies) of open structures in unbounded domains by solving discretised spectral problems, leading to non-Hermitian (possibly quadratic) matrix eigenvalue problems. The study can be extended to highly dispersive materials (metals near the plasmon resonance), leading to nonlinear non-Hermitian eigenvalue problems. In 2018, **we removed the main obstacle to using this approach in highly dispersive media thanks to a universal formulation of the modal expansion in arbitrary dispersive media** (published in *Optics Letters*, December 2018).

Another notable line of work is the treatment of spatial effects in nonlinear electromagnetism. It started with the PhD thesis of Fabien Drouart (spatial solitons in optical fibres). The PhD thesis of Pierre Godard led to a formalism suited to the numerical treatment of scattering problems and to the first finite element models. The ongoing PhD thesis of Jérémy Itier develops this approach so as to treat experimental devices rigorously; I take an active part in this work without being formally its supervisor.

Applications of these models include waveguides (microstructured optical fibres, segmented waveguides), (bi-)gratings, QED cavities, plasmon–soliton coupling, nonlinear plasmonics, metamaterials. . .

Since 2024, I have opened a new research line with E. Chevallier, Associate Professor in the Phyti team (Physics and Image Processing), on the use of differential-geometry methods for processing images from optical experiments.

3 TEACHING

Current courses

Mathematics for Physics (L2 Physics, 2nd year, semester 4)

since 2012

66 h

Lectures: 24 h, tutorials: 30 h, labs: 12 h

Linear algebra (eigenvalues and diagonalisation of matrices), Fourier series, ordinary differential equations, probability and statistics, Python labs

Fundamentals in Optics (M1 Europhotonics, Erasmus Mundus, semester 1)

since 2010

30 h

Lectures: 20 h, tutorials: 10 h

Maxwell's equations, constitutive relations, wave equations, electromagnetic radiation, diffraction, paraxial optics and Gaussian beams (taught in English). Course redesign supported by the A*MIDEX "TIGER" programme for pedagogical transformation.

Ordinary Differential Equations and Great Equations of Physics

(M1 Master in Modelling of Systems and Physical Phenomena, MSPP, Dakar)

since 2020

32 h

Lectures given remotely to students at Université Cheikh Anta Diop (UCAD), Senegal

Ordinary differential equations; the great equations of physics: Laplace and Poisson, heat equation and statistical physics, wave equation, Schrödinger equation and quantum mechanics, Maxwell's equations, continuum mechanics (Navier-Stokes and elasticity)

Formal Tools for Physics (L1, 1st year, semester 2)

since 2024

30 h

Tutorials taught as a flipped classroom

Research supervision (M1/M2)

Master internships and theses

Creation of an international, interdisciplinary Master in Modelling (with Université Cheikh Anta Diop, Dakar, Senegal), 2018

With my colleagues Prof. Serigne Bira Gueye (Université Cheikh Anta Diop, Dakar, Senegal) and Prof. F. Zolla (AMU), I created the Master in Modelling of Systems and Physical Phenomena (MSPP), an interdisciplinary and international AMU/UCAD programme.

Modelling plays an increasingly important role in a wide range of disciplines: the hard sciences, of course, such as physics, chemistry and all the associated engineering sciences, but also biology, medicine, economics, geography, demography and ecology, to name only a few. Beyond the diversity of applications, the same mathematical and numerical techniques are often found at the heart of the models. Moreover, deep-learning techniques have joined the traditional models based on ordinary and partial differential equations or on probabilistic and statistical analysis. The aim of this Master is to train specialists with a broad knowledge of the various types of modelling (with the mathematical and computing skills this requires), able to interact with specialists of a given field (which also requires a broad scientific culture) in order to choose and implement the most suitable modelling techniques. Mastering the geometrical tools of space discretisation and graphical representation is also essential.

History

3.1 International Master Europhotonics

I took part in the working group of the Erasmus Mundus Europhotonics programme *Concepts and Methods for Advanced Optical Imaging from Nanometric to Astronomical Scales*, supported by the Marseille laboratories Institut Fresnel, LAM, PIIM and LP3, which became the International Master Europhotonics.

I also regularly supervise students' group "personal projects".

3.2 Initial training

Since taking up my position at Université Paul Cézanne (Aix-Marseille III) in 1996, I have taught my full teaching load every year (between 192 and 210 hours, tutorial-equivalent, depending on needs). I have been keen to teach at all university levels: undergraduate, professional and research-oriented programmes.

Courses taught previously

2018–2025 (Aix-Marseille Université):

- *Electromagnetism and Optics* (lectures in English 20 h + tutorials 20 h), M1 Fundamental Physics;

2012–2018 (Aix-Marseille Université):

- *Mathematics 3* (lectures 30 h + tutorials 30 h), L2 Physics-Chemistry;

- *Mathematics 4* (lectures 30 h + tutorials 30 h), L2 Physics-Chemistry;

- *Lagrangian and Hamiltonian Formalisms* (lectures 12 h), L2 Physics;

2009–2012 (Université Paul Cézanne):

- *Electromagnetism* (lectures 24 h), L2 Mathematics-Computer Science option;

- *Mathematical Tools I (distributions and Fourier transforms)* (lectures 30 h), L3 Physics;

- *Mathematical Tools II (complex analysis)* (lectures 12 h + 12 h), L3 Physics;

- *Mathematical Tools* (lectures 15 h), L3 Physics-Chemistry;

- *Classical Electromagnetic Theory and Modelling* (lectures 20 h), Research Master in Theoretical Physics. The chapter of *Foundations of Photonic Crystal Fibres* on finite elements in electromagnetism, which I wrote for researchers but starting from minimal prerequisites, serves as the basis for the lecture notes of this course. I was also the Aix-Marseille III correspondent for this programme until the merger of the three Aix-Marseille universities.

Earlier, I also taught the following courses:

- *Physical Thermodynamics* (lectures 18 h), Bachelor in Physical and Chemical Sciences (2nd year);

With Prof. M. Tholomier, I took part in designing the curriculum of the Professional Master in Electromagnetic Compatibility, where I taught:

- *Electromagnetism* (lectures 20 h), Professional Master in Electromagnetic Compatibility;

- *Numerical Methods for Electromagnetism* (lectures 20 h + tutorials 20 h), Professional Master in Electromagnetic Compatibility;

In these courses, I used the software developed in my research (*Gmsh* and *GetDP*, under GNU licence) to introduce students to modelling tools in electromagnetism.

- Tutor for the Computing and Internet Certificate (C2i);

- *Properties of Materials* (lectures 18 h + tutorials 18 h), 1st year of IUP GEII (Electrical Engineering and Industrial Computing);

- *Digital Signal Processing* (lectures 18 h + tutorials 18 h), 1st year of IUP GEII;

- *Numerical Computation in Electromagnetism* (lectures 30 h + tutorials 20 h), DESS GEII (professional postgraduate degree);
- *Classical Electromagnetic Theory and Guided Propagation* (lectures 20 h with F. Zolla, Université de Provence), DEA (research postgraduate degree) in Theoretical Physics, Luminy;
- *Electromagnetism* (lectures 18 h), DEA in Electrical Engineering;
- *Design Methods for Electromagnetic Components and Structures* (lectures 12 h), DEA in Electrical Engineering;
- *Fundamental Physics* (lectures 25 h + tutorials 25 h), Maîtrise (4th year) in Applied Mathematics, Mathematical Engineering option;
- *Applied Mathematics* (lectures 15 h + tutorials 15 h), MST (4th year) Environmental Chemistry; (lectures 15 h + tutorials 15 h), IUP (4th year) Electrical Engineering; (tutorials 10 h), Licence (3rd year) in Physical Sciences;
- *Power Electronics* (lectures 25 h + tutorials 25 h), Licence (3rd year) in Electrical Engineering;
- At ESIM, I was in charge of four electromagnetism courses: I started the course of the ITII PACA Electronics programme and the two courses of the DEA in Electrical Engineering, and I also ran a specialised elective module entitled *Finite Element Method Applied to the Design of Electromagnetic Systems* (6×3.5 h);
- I also created and taught the core course *Numerical Mathematics* for second-year ESIM students;
- From 1987 to 1994, at the University of Liège, I supervised laboratory work in electrical engineering and power electronics in the Applied Electricity group;
- Between 1991 and 1994, at the University of Liège, I created and taught a sixty-hour course on the physics of magnetic materials used in electrical engineering.

3.3 Continuing education

Besides initial training, I have been actively involved in continuing education (mainly in English) on numerical methods in electromagnetism and their application to the design of electrical engineering devices.

- European COMETT programme on electromagnetic modelling, at Katholieke Universiteit Leuven (1991–1994), with the company Infolytica, which designs and sells computer-aided design software for electromagnetic devices. It was intended for European design engineers;
- BEST summer course at the University of Liège (1993) on the finite element method, for engineering students from various countries of the European Community;
- Introductory course on numerical design methods for electromagnetic devices, for the industrial participants of the *Workshop on Electric and Magnetic Fields*, Leuven (1994) and Marseille (1998);
- Continuing-education training *Computer algebra and Mathematica* (1995) for teachers of the *classes préparatoires*;
- I taught an electromagnetism course in the ITII PACA Electronics programme (1995–2003) for technicians training to become engineers in a work-study scheme.

4 RESEARCH

My first research on the modelling of electromagnetic fields (magnetostatics and eddy currents) was carried out in the Applied Electricity group of the University of Liège, headed by Professor W. Legros, from October 1984 to November 1994.

Then, in 1995 and 1996 at ESIM, I started a modelling activity on electromagnetic systems in collaboration with the team of Professor M. Tholomier (LaGEM) of Université d'Aix-Marseille III, within Cegema (Centre de Génie Électrique de Marseille-Aix, bringing together LaGEM, ENSAM Aix-en-Provence and the EEPs department of ESIM), taking into account the regional scientific and industrial environment. Following an industrial contract, I became interested in modelling parasitic capacitances in inductive components. I pursued this activity at Université d'Aix-Marseille III when I joined LaGEM in September 1996, and it was developed by C. Millard in her PhD thesis *Modelling of parasitic capacitances of wound components*, using the Partial Element Equivalent Circuit method to obtain a simple model validated against experimental results provided by Gemplus and LETI.

Since 2000, I have been a member of Institut Fresnel, CNRS joint research unit UMR 7249 (formerly 6133, CLARTE and SEMO teams).

My main research activity concerns electromagnetic scattering and guided propagation, with applications such as microstructured optical fibres, metamaterials and photonic crystals, and nonlinear optics.

I built on my experience of the finite element method at low frequency to apply it to modelling in optics (where it was hardly used) and microwaves, edge elements and perfectly matched layers (PMLs) allowing an efficient transposition to propagation problems. Finite elements bring flexibility in the treatment of boundary conditions (for instance, the periodic problems of diffraction gratings and photonic crystals), the representation of arbitrary geometries, and the treatment of material inhomogeneity and anisotropy (for instance, for invisibility cloaks, nonlinear optics and gyromagnetic materials).

I took up the covariant formulation of electromagnetism based on differential geometry (exterior calculus), which is also the natural framework for the theory of edge elements, to show that any change of coordinates, however exotic, is exactly equivalent to a modification of the material properties (most often permittivity and permeability). This unifies a wide variety of methods, such as PMLs, which correspond to a complex-valued change of coordinates, helicoidal coordinates, which allow a two-dimensional treatment of twisted problems, and invisibility cloaks and negative-index materials, which correspond respectively to puncturing and folding space.

The following work has been carried out along these lines:

- **Waveguides and microstructured optical fibres.**
 - From 2000, I developed with F. Zolla and S. Guenneau a numerical model for optical fibres. It is a vector model based on the study of a cross-section of the guide, using edge finite elements for the transverse field and nodal elements for the longitudinal component. Propagation modes are obtained by solving eigenvalue problems numerically.
 - The model was developed with the *GetDP* software, originally designed by P. Dular^{†3}, FNRS researcher, and C. Geuzaine, University of Liège (Belgium), with whom I actively collaborate. C. Geuzaine, then a postdoctoral fellow at Caltech, spent a month as visiting professor at Institut Fresnel in July 2005. During this period, we improved the eigenvalue computation in *GetDP* (coupling with the ARPACK library and development of an algorithm for quadratic eigenvalue problems).
 - This model was extended by Y. Ould Agha during his PhD to helicoidal fibres (keeping an exact two-dimensional approach thanks to a covariant formulation of Maxwell's equations in helicoidal coordinates), to study a spontaneous twisting phenomenon that hindered the industrial production of microstructured fibres. This research continued in collaboration with the Department of Applied Mathematics of the University of Liverpool on asymptotic approximations. Professor A. B. Movchan, head of this department, and I obtained a British Council Alliance grant for 2005 and 2006, which supported our joint research through regular reciprocal visits.

³The symbol [†] denotes a deceased person: Patrick Dular passed away on 6 September 2017 and Mireille Commandré on 11 July 2017.

- With G. Renversez, then Associate Professor at Université d’Aix-Marseille III, and F. Drouart, then a PhD student, we studied nonlinearities in microstructured optical fibres; a paper entitled *A finite element analysis of spatial solitons in optical fibres* was published in COMPEL. This research continued in collaboration with Professor A. Ferrando, Department of Optics, University of Valencia (Spain).
- With F. Zolla and Y. Ould Agha (PhD, November 2007), we studied the use of PMLs for computing leaky modes in open waveguides. This problem, now well understood and solved, was the last element needed to complete our two-dimensional model of an ideal electromagnetic waveguide. We also extended this technique to losses in helicoidal fibres.
- I co-authored the book *Foundations of Photonic Crystal Fibres*, published in 2005 by Imperial College Press (with F. Zolla, G. Renversez, B. Kuhlmeier, S. Guenneau and D. Felbacq). It presents the state of the art (as of 2004) in the modelling of microstructured fibres while laying solid theoretical foundations for future modelling.
- Microstructured fibres also led to a DGA contract (no. 05.34.053, 30 months, total 163,288 euros), *New numerical approaches for the modelling of microstructured fibres*, with G. Renversez as principal investigator for Institut Fresnel. It provided modelling support to the experimentalists of the Glass and Ceramics Laboratory in Rennes and to the PERFOS fibre fabrication platform in Lannion, and funded a postdoctoral researcher (Ion Vasilief) for 18 months.

- **Metamaterials, photonic crystals and transformation optics.**

- Photonic metamaterials have grown considerably, and the finite element models we developed proved very useful for the various numerical problems arising in this context. We adapted the guided-propagation model to electromagnetic Floquet–Bloch waves in periodic structures to compute dispersion curves. For instance, we studied arrays of “Swiss roll” structures proposed by Prof. John Pendry (Imperial College London) to build metamaterials with exotic optical properties.
- Institut Fresnel (CLARTE team) was a partner of the ANR project MAGNETO-PHOT (BLAN06-2-135594) on gyromagnetic photonic crystals (coordinator: Pierre Beauvillain, IEF; other partners: IEF, GEMAC, FEMTO-ST; October 2006 – October 2009; total 447,400 euros including 94,610 euros for CLARTE). F. Zolla, S. Guenneau and I were in charge of the finite element modelling of gyromagnetic structures.
- Institut Fresnel (CLARTE team) was a partner of the ANR project POEM (06-NANO-008-03, Nanosciences and Nanotechnologies programme 2006) on electromagnetic wave propagation in metamaterials (coordinator: Didier Felbacq, GES; other partners: GES Montpellier II, ANAM Toulon, LASMEA Clermont-Ferrand, FEMTO-ST Besançon; October 2006 – October 2009; total 278,200 euros including 82,100 euros for CLARTE). F. Zolla, S. Guenneau and I were in charge of the finite element modelling of both the auxiliary problems of homogenisation theory and the homogenised scattering problems. A new type of problem related to the homogenisation of “wire crystals” was also studied.
- With F. Zolla and S. Guenneau, we undertook the study of invisibility cloaks. The idea was proposed in 2006 by Prof. J. Pendry (Imperial College London) as one of the most spectacular possible applications of photonic metamaterials. We published a paper with John Pendry in *Optics Letters* on the modelling of two-dimensional cloaks, which attracted public interest and wide media coverage (see Section 4.4.6).
- This research on metamaterials and transformation optics continued within the ANR project OPTRANS (2010–2014), with GES (Montpellier), Institut Fresnel (Marseille), the IMATH mathematics laboratory (Toulon) and Institut Langevin (Paris); total funding 502,070 euros including 233,542 euros for Institut Fresnel, which funded among other things a postdoctoral position. The research addresses the design of optical functions with metamaterials. A first part uses transformation optics: how to build, from a ray tracing and/or wavefronts, an equivalent geometrical transformation yielding the equivalent materials. A second part, called “dehomogenisation”, solves the inverse problem of homogenisation, i.e. computes the characteristics of a metamaterial whose homogenisation in the working frequency band yields the material properties obtained in the first step.

- **Electromagnetic scattering.**

- In 2002, I modelled microwave devices in a collaboration with Institut Français du Pétrole (Rueil-Malmaison) on a multiphase microwave flowmeter (DESS internship of Aurélie Laville in 2002, a 44,000-euro contract for modelling the device, DEA internship of Adèle Nguyen in 2004).

- In December 2005, I started a collaboration with Agfa-Gevaert (Antwerp, Belgium) on light scattering by inkjet printer pigments (random distributions of angular micro-crystals with a very high and strongly anisotropic refractive index). A professional Master internship (Georgia Diamantopoulou) and a feasibility study were carried out on a two-dimensional model between March and June 2006. In 2007, within Agfa's *Hypercure* project for next-generation inkjet printers, a collaboration with Agfa-Gevaert (140,000 euros) allowed us to host Professor Min Zhang from Xidian University (China) in our laboratory.
- I collaborated with the MAP2 team of Institut Fresnel, headed by Prof. M. Commandré[†] (École Centrale de Marseille), on a contract with STMicroelectronics (Rousset) to model the optical part of a CMOS image sensor pixel. With F. Zolla, I provided part of the scientific supervision (electromagnetic optics, diffraction gratings and finite elements) of the numerical modelling work of Guillaume Demésy, CIFRE PhD student. This collaboration continued with the *Spectrogaz* project (2011–2014), which funded among other things the PhD of Benjamin Vial, where grating models are now examined from a spectral point of view. The difficulty is that in these open problems there are generally no modes in the strict sense, but one can define quasi-modes (similar to the leaky modes of waveguides) with complex frequencies, and the continuous spectrum also plays an important role. PMLs are a fundamental tool well suited to this study.
- With H. Tortel, Associate Professor at Aix-Marseille Université (Institut Fresnel), I initiated the development of a finite element code for three-dimensional structures optimised for inverse problems in electromagnetism.
- I also took part in ERT51-LaCEM, created by Prof. Tholomier (with Institut Fresnel as scientific laboratory and the Eurocopter group as industrial partner), devoted to electromagnetic compatibility in aeronautical systems.

- **Nonlinear optics.**

Nonlinear optics offers many possibilities for new optical devices. Besides the PhD thesis of Fabien Drouart on spatial solitons in microstructured optical fibres, I supervised with F. Zolla the study of scattering by materials with a nonlinear refractive index in the PhD thesis of Pierre Godard. I take an active part in the research carried out in the PhD thesis of Jérémy Itier, which follows on from Pierre Godard's work and is expected to be defended in 2027.

- **Quantum optics.**

I collaborated with Prof. Thomas Durt (École Centrale de Marseille since September 2010), a specialist in quantum mechanics, in particular its foundations and quantum information, on problems at the boundary between classical electromagnetism based on Maxwell's equations and quantum problems requiring the notion of photon, a notion notoriously poorly understood. We co-supervised the Master 2 internship (Sciences de la matière, physics track, École Normale Supérieure de Lyon) of Vincent Debierre, *The role of time in quantum mechanics – Classical-quantum transition*. It appeared that the methodologies of the “quantum” approach and of the “electromagnetic modelling” approach are very different, but that confronting them would be mutually enriching and lead to interesting findings. For instance, the use of PMLs in simple models of the Schrödinger equation in open domains turned out to be little known in the literature. Vincent Debierre then obtained a fully CNRS-funded PhD grant, co-supervised by Prof. Durt and myself, from September 2012 to September 2015, on boundaries and transitions in quantum mechanics. This work led to several journal publications.

- **Modelling of QED cavities.**

With G. Demésy, we studied the electromagnetic modelling of the **QED cavity of the Haroche** (Nobel Prize in Physics 2012)–Raimond–Brune team at LKB, to show that the photon lifetime is bounded by the characteristics of the leaky mode in which the photons “live”. Computable with finite elements and PMLs, this approach estimates the lifetime by an apparently purely classical computation, which nevertheless lies at the boundary between classical and quantum electrodynamics. This research continued in collaboration with Prof. C. Geuzaine, University of Liège (PhD of N. Marsic, supervised by Prof. Geuzaine, defended in January 2016), extending the finite element method to high-order elements for both interpolation and geometrical representation, which is needed to represent the cavity mirrors accurately enough.

- **Segmented waveguides.**

Combining our know-how in modelling waveguides and periodic structures (PMLs, Floquet–Bloch conditions and quasi-modal analysis), I proposed with G. Demésy and F. Zolla novel structures for guiding electromagnetic waves. They consist of chains of identical, non-touching elements, and our model selects the quasi-modes whose leakage is low enough for effective guiding. This work was the subject of an invited talk at the IEEE CAMA conference, Juan-les-Pins (France), in November 2014.

- **Modal analysis of dispersive media.**

One of the main lines of my recent work is the modal analysis of open problems. Combining finite elements and PMLs, one can determine the “leaky” resonances (with complex frequencies) of open structures in unbounded domains by solving discretised spectral problems, leading to non-Hermitian (possibly quadratic) matrix eigenvalue problems. The study can be extended to highly dispersive materials (metals near the plasmon resonance), leading to nonlinear non-Hermitian eigenvalue problems.

To develop this research, the ANR project RESONANCE (ANR-16-CE24-0013, *Theory and numerical modeling of optical resonance*) on **quasi-modal analysis**, of which I was the local principal investigator (budget 367,200 euros including 146,880 euros for Institut Fresnel, with the funding of a PhD co-supervised with Ph. Lalanne), was accepted in 2016 (other partners: LP2N, Institut d’Optique d’Aquitaine (Ph. Lalanne), and Institut d’Optique, Palaiseau (C. Sauvan)).

In 2018, we removed the main obstacle to using this approach in highly dispersive media thanks to a universal formulation of the modal expansion in arbitrary dispersive media (published in *Optics Letters*, December 2018).

We also collaborate with Jose E. Roman (Universitat Politècnica de València), who develops the SLEPc library for numerical eigenvalue computations and uses our models as test cases for his library, now the reference in the field, and with C. Geuzaine, since our models are integrated into the open-source ONELAB platform that he develops.

- **Information geometry.**

Since October 2024, I have opened a new research line with E. Chevallier, Associate Professor in the Phyti team (Physics and Image Processing), on the use of differential-geometry methods for processing images from optical experiments. I supervise, with E. Chevallier as co-supervisor, the PhD thesis of Gabriel Trindade, *Extrinsic statistics for polarisation matrices of electromagnetic waves*. This multidisciplinary thesis involves differential geometry, image processing and optics. A first journal paper appeared in *Information Geometry* in 2026.

4.1 Contracts, industrial and other activities

- As a research engineer at the University of Liège (1984–1994), I worked on several industrial contracts: eddy-current losses in foil-wound transformer windings for France-Transfo (Metz); dielectric strength computations for CDC, Alcatel Câbles (Charleroi, Dunkirk) and Balteau Transformateurs de Mesure (Beyne-Heusay); a magnetoforming device for Athaena; electromagnetic devices (guiding and heating) of a steel-sheet production line for the Cockerill-Sambre Metallurgical Research Centre (Liège); a linear electromagnetic actuator for vacuum circuit breakers for EIB (Dison).
- Principal investigator of the CREMSI project *Development of a numerical model for the characterisation of inductors at high frequency and the determination of parasitic capacitance*, with ESIM and Gemplus (1998).
- European project PHPC-PACA, *Electromagnetic compatibility action: modelling the susceptibility of a printed circuit board in its casing*, with Simulog and SIEMI (1999).
- Principal investigator of a project of the French Ministry of Foreign Affairs for cooperation with South Africa, *Optimisation of thermal losses in an inductor for on-board power electronics*, with Rand Afrikaans University (Prof. W. Cronje) (2000–2001, 100,000 FRF).
- Principal investigator of a research collaboration contract with Saint-Gobain Recherche (with F. Zolla), *Interaction between infrared radiation and a glass cylinder of circular cross-section with metallic cladding* (April–August 2001, 100,000 FRF).
- Principal investigator of a contract with Institut Français du Pétrole (no. 28286), *Finite element electromagnetic modelling of a sensor for a multiphase flowmeter* (June 2003 – June 2004, 44,000 euros).
- Principal investigator of a British Council Alliance project (2005 and 2006, 3,000 euros per year for each team) with Prof. A.B. Movchan, Department of Applied Mathematics, University of Liverpool, on photonic and phononic crystals.
- Principal investigator of a contract with Agfa-Gevaert (Antwerp, Belgium) on the optical response of printer inks: *Simulation method for the calculation of the optical response of densely packed pigment layers by the finite element method*. A feasibility study (6,000 euros) and a paid professional Master internship took place from March to June 2006.
- Principal investigator of a contract with Agfa-Gevaert (Antwerp, Belgium) within its *Hypercure* project for next-generation inkjet printers: a two-year collaboration (2007–2008, 140,000 euros), with the recruitment of Prof. Min Zhang (Xidian University, China) to develop new numerical models of two- and three-dimensional random media.
- Collaboration with the MAP2 team of Institut Fresnel, headed by Prof. M. Commandré[†], on a contract with STMicroelectronics (Rousset), modelling the optical part of a CMOS image sensor pixel.
- **ANR MAGNETO-PHOT (BLAN06-2-135594)**, 2006–2009: Institut Fresnel (CLARTE team) was a partner of this project on gyromagnetic photonic crystals (coordinator: Pierre Beauvillain, IEF; other partners: GEMAC, FEMTO-ST; total 447,400 euros including 94,610 euros for CLARTE). F. Zolla, S. Guenneau and I were in charge of the finite element modelling of gyromagnetic structures.
- **ANR POEM (06-NANO-008-03)**, 2006–2009: Institut Fresnel (CLARTE team) was a partner of this project (Nanosciences and Nanotechnologies programme 2006) on electromagnetic wave propagation in metamaterials (coordinator: Didier Felbacq, GES; other partners: ANAM Toulon, LASMEA Clermont-Ferrand, FEMTO-ST Besançon; total 278,200 euros including 82,100 euros for CLARTE). F. Zolla, S. Guenneau and I were in charge of the finite element modelling of both the auxiliary problems of homogenisation theory and the scattering problems.
- **ANR OPTRANS (10-BLAN-1024-02)**, 2010–2014: partners GES (Montpellier), Institut Fresnel (Marseille),

IMATH (Toulon) and Institut Langevin (Paris); total funding 502,070 euros including 233,542 euros for Institut Fresnel, which funded among other things a postdoctoral position.

- *Spectrogaz* project (2011–2014), funded by FUI and the PACA Region: collaboration between BERTIN, SILIOS TECHNOLOGIES and Institut Fresnel. The project funded among other things the PhD of Benjamin Vial.
- **ANR RESONANCE (ANR-16-CE24-0013)**, 2016–2020: *Theory and numerical modeling of optical resonance, on quasi-modal analysis*. Local principal investigator (budget 367,200 euros including 146,880 euros for Institut Fresnel, with the funding of a PhD co-supervised with Ph. Lalanne). Other partners: LP2N, Institut d'Optique d'Aquitaine (Ph. Lalanne) and Institut d'Optique, Palaiseau (C. Sauvan).
- **Industrial contract with CERSA-MCI (Gardanne) – Power scattered by an air bubble in an optical fibre (2019)**: CERSA-MCI is an SME that designs and manufactures instruments for measuring and controlling optical fibre production. The aim was to provide a dedicated modelling tool estimating the power scattered by an air bubble in an optical fibre, in order to design a real-time optical detector of manufacturing defects. The model combines diffraction (the bubble is a few wavelengths in size) and geometrical optics (the optical system of the sensor). A 12,000-euro contract managed by PROTISVALOR (technology-transfer subsidiary of Aix-Marseille Université).
- **Senior Member, Advisory Board – GHOST SPACE SRL (Liège, Belgium), since August 2025**: a deep-tech spin-off of the University of Liège supported by the ERC and the European Space Agency. Strategic advice on a technology for characterising and correcting stray light, based on time-of-flight measurements with an ultrafast camera, for space-borne imaging. Scientific collaboration with the founder and CEO, Prof. L. Clermont, for about fifteen years; L. Clermont gave a seminar on this technology at Institut Fresnel at my invitation (November 2023).

4.2 Supervision

4.2.1 PhD theses and current positions of former students

- Co-supervisor (70%, with W. Legros) of the PhD thesis of P. Dular[†] (FNRS research fellow), *Modelling of the magnetic field and eddy currents in three-dimensional nonlinear systems*, May 1994, University of Liège;
FNRS Research Director, Belgium
- Co-supervisor (70%, with W. Legros) of the PhD thesis of F. Delincé (FNRS research fellow), *Modelling of transient regimes in systems including nonlinear and hysteretic magnetic materials*, September 1994, University of Liège;
Engineer at ABB Jumet, Belgium
- Co-supervisor (70%, with W. Legros) of the PhD thesis of J.-F. Remacle, *Error estimation in finite element modelling of electromagnetic fields: application to mesh optimisation*, June 1997, University of Liège;
Professor, Université catholique de Louvain (UCLouvain), Belgium
- Supervisor (100%) of the PhD thesis of C. Millard, *Modelling of parasitic capacitances of wound components*, October 1996 – October 1999, Université d'Aix-Marseille III;
Manager in industry
- Supervisor (50%, with F. Zolla) of the PhD thesis of S. Guenneau, *Homogenisation of quasicrystals and mode analysis in photonic crystal fibres*, April 2001, Université d'Aix-Marseille III;
Professor, Imperial College London, United Kingdom
- Supervisor (50%, with F. Zolla) of the PhD thesis of Y. Ould Agha, *Real and complex geometrical transformations: application to leaky modes in microstructured optical fibres*, November 2004 – November 2007, Aix-Marseille Université;
Associate Professor, PhLAM laboratory, Université de Lille
- Supervisor (50%, with G. Renversez) of the PhD thesis of F. Drouart (October 2005 – November 2008) on the finite element modelling of nonlinearities in microstructured optical fibres;
Engineer at Eutelsat Group
- Supervisor (50%, with F. Zolla) of the PhD thesis of P. Godard (September 2006 – September 2009) on the modelling of guided propagation in nonlinear optics;
Associate Professor, Université de Poitiers
- Supervisor (50%, with H. Tortel) of the PhD thesis of M. Malidé (December 2006 – September 2010) on the finite element modelling of three-dimensional structures for inverse problems in electromagnetism;
Teacher, Comoros
- Supervisor (40%, with F. Zolla and M. Commandré[†]; CIFRE thesis with STMicroelectronics) of the PhD thesis of Guillaume Demésy (April 2006 – March 2009), *Three-dimensional electromagnetic modelling of complex gratings. Application to spectral filtering in CMOS image sensors*;
Associate Professor, Institut Fresnel, Aix-Marseille Université
- Supervisor (40%, with F. Zolla and M. Commandré[†]; CIFRE thesis with SILIOS) of the PhD thesis of Benjamin Vial (April 2010 – April 2013), *Development of infrared multispectral filters based on subwavelength structured materials*;
Researcher, Imperial College London, United Kingdom
- Supervisor (50%, with T. Durt) of the PhD thesis of Vincent Debierre, fully funded by a CNRS grant (September 2012 – September 2015), on boundaries and transitions in quantum mechanics, *The photon wave function, in principle and in practice*;

Researcher, Max Planck Institute for Nuclear Physics, Heidelberg, Germany

- Supervisor (70%, with Ph. Lalanne) of the PhD thesis of Minh Duy Truong, *Theory and numerical modeling of photonic resonances*, funded by ANR RESONANCE (November 2016 – November 2020);

Researcher, University of Twente, The Netherlands

- Supervisor (40%, with E. Chevallier, Associate Professor, as co-supervisor) of the PhD thesis of Gabriel Trindade, *Extrinsic statistics for polarisation matrices of electromagnetic waves* (October 2024 – end of 2027);
Thesis in progress
- Mentor for the *Habilitation à Diriger des Recherches* of G. Demésy (2026, submission in progress).

4.2.2 Master's level (DEA, DESS, Master 2)

- DEA internship of Sylvain Ferrand: *Defect analysis in prestressed cables by reflectometry*, DEA in Electrical Engineering, Aix-Marseille, 1995;
- DEA internship of Michel Salatti: *Modelling of hysteresis in magnetic materials*, DEA in Electrical Engineering, Aix-Marseille, 1996;
- DESS internship of Aurélie Laville: *Study of a multiphase microwave flowmeter*, DESS GEII, Aix-Marseille III, 2002;
- DEA internship (Optics and Image Processing) of Frédéric Bordas (with G. Renversez): *Properties of high-index microstructured optical fibres*, March–June 2004;
- DEA internship (Optics and Image Processing) of Adèle Nguyen (with F. Zolla): *Radiation pattern of the antenna of a microwave flowmeter with a three-dimensional finite element model*, March–June 2004;
- Professional Master 2 internship (Mechanical Engineering and Scientific Computing) of Sébastien Bohigues (with F. Zolla): *Leaky modes in microstructured optical fibres*, April–August 2005;
- Research Master 2 internship (Optics and Image Processing) of Fabien Drouart (with G. Renversez): *Towards the study of nonlinearity in microstructured fibres*, March–June 2005;
- Professional Master 2 internship (Electromagnetic Compatibility) of Georgia Diamantopoulou: *Testing the simulation method for the calculation of the optical response of densely packed pigment layers by the finite element method*, March–June 2006;
- Research Master 2 internship (Acoustics) of Bastien Rizzon (with F. Zolla): *Three-dimensional study of invisibility cloaks*, March–August 2007;
- Master 2 internship (Sciences de la matière, physics track, École Normale Supérieure de Lyon) of Vincent Debierre (with T. Durt and F. Zolla): *The role of time in quantum mechanics – Classical-quantum transition*;
- MSPP Master internship (AMU/UCAD) of Massar Diadhiou: *Two-dimensional numerical modelling of the wave equation*, March–July 2022;
- MSPP Master internship (AMU/UCAD) of Fallou Mbacké Lô: *Simulation of the field diffracted by an aperture in a plane screen with the angular spectrum method implemented in Python*, March–July 2023;
- MSPP Master internship (AMU/UCAD) of Dieynaba Sanou Mbaye (with F. Zolla, at Institut Fresnel, Marseille): *Modelling biodiversity: non-transitive interactions in Escherichia coli populations and the influence of the interaction radius*, March–July 2023.

4.2.3 Engineering final-year projects

As a research engineer at the University of Liège, I provided the scientific supervision of the following engineering final-year projects (Faculty of Applied Sciences):

- *Numerical modelling of the dynamic behaviour of a contactor*, Pierre Thiry, 1987–88, Electrical-Mechanical Engineering, Electricity option;
- *Determination of the distribution of magnetic field lines in saturated media*, Bernard Delvaux, 1987–88, Electrical-Mechanical Engineering, Electricity option;
- *Modelling of the electromagnetic field by surface integrals with second-order elements*, François Delincé, 1988–89, Mechanical Engineering, Engineering Physics option;

- *Modelling of an electromagnetic circuit taking magnetostriction into account*, Jean-Michel Gillard, 1989–90, Mechanical Engineering, Engineering Physics option;
- *Three-dimensional magnetic field generated by an arbitrary structure*, Patrick Dular[†], 1989–90, Electrical Engineering, Power Systems option;
- *Determination of eddy currents by the direct surface integral method*, Mathias Grenacs, 1989–90, Electromechanical Engineering, Aerospace option;
- *Modelling of the hysteresis loop of magnetic materials*, François Henrotte, 1990–91, Mechanical Engineering, Engineering Physics option;
- *Application of the boundary element method to conducting materials in axisymmetric structures*, Luc Defrère, 1991–92, Mechanical Engineering, Engineering Physics option;
- *Study of an electromagnetic launcher*, Christian Servais, 1992–93, Electrical Engineering (Physical Electronics).

4.3 Conferences

I presented papers at the following conferences:

- Modelec, La Grande Motte (France), October 84;
- International Symposium on AI, Expert Systems and Languages in Modelling and Simulation, Barcelona (Spain), June 87;
- International Conference on Electrical Machines, Pisa (Italy), September 88;
- Intermag 89, Washington (U.S.A.), March 89;
- Betech 90, Newark (U.S.A.), July 90;
- MMM 90, San Diego (U.S.A.), October 90;
- RRES 91, Milan (Italy), April 91;
- 13th BEMIC, Tulsa (U.S.A.), August 91;
- Compumag 91, Sorrento (Italy), July 91;
- EURISCON 91, Corfu (Greece), June 91;
- Numelec 92, Grenoble (France), March 92;
- CEFC 92, Claremont (USA), August 92;
- 14th BEMIC, Seville (Spain), November 92;
- Colloque 'Méthodes informatiques de la conception industrielle', Marseille (France), June 93;
- Compumag 93, Miami (U.S.A.), November 93;
- MMM 93, Minneapolis (U.S.A.), November 93;
- ACES 94, Monterey (U.S.A.), March 94;
- CEFC 94, Aix-les-Bains (France), July 94;
- 16th BEMIC, Southampton (UK), July 94;
- Compumag 95, Berlin (Germany), July 95;
- Betech 95, Liège (Belgium), September 95;
- EMF 96, Liège (Belgium), May 96;
- EPNC 96, Poznan (Poland), June 96;
- Numelec 97, Lyon (France), March 97;
- EPNC 98, Liège (Belgium), September 98;
- EMF 98, Marseille (France), May 98;
- Numelec 2000, Poitiers (France), April 2000;
- EMF 2000, Ghent (Belgium), May 2000;
- PIERS 2000, Boston (USA), July 2000;
- Compumag 2001, Evian (France), July 2001;
- ACOMEN 2002, Liège (Belgium), May 2002;
- EPNC 2002, Leuven (Belgium), July 2002;
- IUTAM, Liverpool (UK), July 2002;
- ISEM 2003, Versailles (France), May 2003;
- EMF 2003, Aachen (Germany), September 2003;
- Numelec 2003, Toulouse (France), November 2003;
- EPNC 2004, Poznan (Poland), June 2004;
- Mathematics 2005, Liverpool (UK), April 2005;
- Acomen 2005, Ghent (Belgium), May 2005;
- CEM 2006, Aachen (Germany), April 2006;
- ACE'06, Boston (USA), May 2006;
- EPNC 2006, Maribor (Slovenia), June 2006;

- EMF 2006, Aussois (France), June 2006;
- Etopim 7, Sydney (Australia), July 2006;
- Numelec 2006, Lille (France), November 2006;
- Compumag 2007, Aachen (Germany), June 2007;
- ACE'07, Aachen (Germany), June 2007;
- PIERS 2007, Prague (Czech Republic), August 2007;
- ISEF 2007, Prague (Czech Republic), September 2007;
- Meta'08, Marrakech (Morocco), May 2008;
- Acomen 2008, Liège (Belgium), May 2008;
- Esco'08, Jetrichovice (Czech Republic), June 2008;
- EPNC 2008, Lille (France), July 2008;
- Prague Photonics, Prague (Czech Republic), August 2008;
- Numelec 2008, Liège (Belgium), December 2008;
- SPIE, Prague (Czech Republic), April 2009;
- EMF 2009, Mondovì, (Italy), May 2009;
- Etopim 8, Rethymnon, (Greece), June 2009;
- Tacona Photonics 2009, Bad Honnef, (Germany), October 2009;
- Compumag 2009, Florianopolis, (Brazil), November 2009;
- PIERS 2010, Xi'an, (China), March 2010;
- GDR PhoNomi2, Brussels (Belgium), May 2010;
- EPNC 2010, Essen, (Germany), June 2010;
- Tacona Photonics 2010, Bad Honnef, (Germany), November 2010;
- CNM 2011, Paris (France), March 2011;
- PIERS 2011, Marrakesh (Morocco), March 2011;
- SPIE 2011, Prague (Czech Republic), April 2011;
- Tacona Photonics 2011, Bad Honnef, (Germany), October 2011;
- Acomen 2011, Liège (Belgium), November 2011;
- PIERS 2012, Kuala Lumpur (Malaysia), March 2012;
- Numélec 2012, Marseille (France), July 2012;
- ICEAA'12, Cape Town (South Africa), September 2012;
- Tacona Photonics 2012, Bad Honnef, (Germany), October 2012;
- EMF 2013, Bruges (Belgium), April 2013;
- Compumag 2013, Budapest(Hungary), July 2013.
- CNM 2013, Paris (France), October 2013;
- URSI Radio 2014, Mauritius, April 2014;
- Acomen 2014, Ghent (Belgium), June 2014;
- IEEE CAMA, Juan-les-Pins (France), November 2014.
- Numélec 2015, Saint-Nazaire (France), June 2015;
- PIERS 2015, Prague (Czech Republic), July 2015;
- 2nd Workshop on Seismic Metamaterials 2016, Dakar (Senegal), May 2016;
- Modelling and Simulation for Nanophotonics, Sophia Antipolis (France), October 2016;
- Waves in periodic media and metamaterials, Cargèse (France), November 2016;
- 'Mathematical and Numerical Modeling in Optics' Workshop of the IMA (Institute for Mathematics and its Applications), Minneapolis (USA), December 2016 ;
- SPIE 2017, Prague (Czech Republic), April 2017;
- Numélec 2017, Paris (France), November 2017;

- GDR Ondes "Advanced theoretical and numerical methods for waves in structured media", Paris (France), March 2018;
- EMF 2018, Darmstadt (Germany), April 2018;
- European Workshop on Acoustic & Mechanical Metamaterials, London (UK), July 2018;
- SPIE 2019, Prague (Czech Republic), April 2019;
- PIERS 2019, Rome (Italy), June 2019;
- Compumag 2019, Paris (France), July 2019;
- Berlin TNTN2020, Berlin (Germany), February 2020;
- META 2023, Paris (France), July 2023;
- Numélec 2024 (Toulouse), July 2024.

4.4 Publications

4.4.1 Books

Author:

1. Foundations of Photonic Crystal Fibres (F. Zolla, G. Renversez, **A. Nicolet**, B. Kuhlmeiy, S. Guenneau and D. Felbacq), Imperial College Press, London, January 2005, ISBN 1-86094-507-4.
2. Foundations of Photonic Crystal Fibres (Frédéric Zolla, Gilles Renversez, **André Nicolet**, Boris Kuhlmeiy, Sébastien Guenneau, Didier Felbacq, Alexander Argyros, Sergio Leon-Saval) Imperial College Press, London, **Second edition**, August 2012, ISBN-13: 978-1-84816-728-5, ISBN-10: 1-84816-728-8.
Augmented and revised edition : 343pp. → 511pp.

Editor:

1. Electric and Magnetic Fields : From Numerical Models to Industrial Applications (**A. Nicolet**, R. Belmans editors), Plenum Publishing Corporation, New York, May 1995, ISBN 0306449919.
2. Boundary Element Technology X (M.H. Aliabadi, C.A. Brebbia, P. Dular[†], **A. Nicolet** editors), Computational Mechanics Publications, 1995, ISBN 1 85312 317 X.

Book chapters:

1. Metamaterials and Plasmonics: Fundamentals, Modelling, Applications (Saïd Zouhdi, Ari Sihvola, and Alexey Vinogradov, Editors), Geometrical Transformations for Numerical Modelling and for New Material Design in Photonics (in Part II, Transformation Media and Cloaking), **André Nicolet**, Frédéric Zolla, Yacoub Ould Agha, and Sébastien Guenneau, pp. 49–62, Springer-Verlag Berlin Heidelberg New York, 2009, ISBN 978-1-4020-9406-4.
2. Photonics for Safety and Security (Antonello Cutolo, Anna Grazia Mignani, Antonella Tajani, Editors), Metamaterials and the Mathematical Science of Invisibility (A. Diatta, S. Guenneau, **A. Nicolet**, and F. Zolla), World Scientific, Singapore, 2013, ISBN-13: 9789814412964.
3. Gratings: Theory and Numeric Applications, Editor: Evgeny Popov, E-book, ISBN:2-85399-860-4 (2013): G. Demésy, F. Zolla, **A. Nicolet**, and B. Vial, Chapter 5: Finite Element Method.
4. Felbacq, D., F. Zolla et al. Metamaterials : theory and Applications. Pan Stanford. New Jersey London. First Edition (2017): General Introduction (Didier Felbacq, **André Nicolet**, and Frédéric Zolla), Transformation Optics in a Nutshell (**André Nicolet**), Scattering Problems: Numerical Methods (FEM, Multiple Scattering) (Didier Felbacq, **André Nicolet**, and Frédéric Zolla), Stiff Problems: High Contrast Objects (Didier Felbacq, **André Nicolet**, Frédéric Zolla, and Guy Bouchitté).

5. **Chapter 1: Maxwell's equations, constitutive relations and classification of electromagnetic media (André Nicolet).**

In: *Optics and Photonics I: Foundations* (World Scientific Publishing), vol. 4 of *Compendium on Electromagnetic Analysis: From Electrostatics to Photonics: Fundamentals and Applications for Physicists and Engineers*. Vadim Markel (volume Editor), Igor Tsukerman (Editor-in-Chief). pp. 1–59, 2020. World Scientific. ISBN 978-981-3270-31-2.

Invited by Vadim Markel, I wrote the introductory chapter (59 pages) of volume 4 of this five-volume encyclopedia. The chapter offers a modern overview of the structure of electrodynamics. The first part presents Maxwell's equations in a geometric (premetric) formulation, independent of any underlying medium, introducing the notions at an elementary level so that the text is self-contained. The second part develops a modern macroscopic theory of electromagnetic media based on charge-current potentials defined independently of any microscopic model, and proposes a systematic and exhaustive classification of constitutive relations. The chapter concludes with a presentation of transformation optics, based on the equivalence between geometrical transformations and constitutive relations, and serves as a reference for the classification of electromagnetic materials.

4.4.2 Theses

1. "Contribution à la modélisation des systèmes électromagnétiques" (**A. Nicolet**). *Habilitation à Diriger des Recherches* (63rd section), January 1996, Université d'Aix-Marseille III.
2. "Modélisation du champ magnétique dans les systèmes comprenant des milieux non linéaires" (**A. Nicolet**). PhD thesis, University of Liège, November 1991, Faculty of Applied Sciences, publication series no. 139.
3. "Etude de la caractérisation d'un bolus de radio isotope de courte demi-vie par ses paramètres temporels" (**A. Nicolet**). Engineering final-year project, University of Liège, 1984.

4.4.3 Journal articles - Google Scholar h-index = 32, citations > 3900

1. Numerical modelization of transient current in relay (W. Legros, **A. Nicolet**, M. Paganini), *IEEE Transactions on Magnetics*, vol. 25, no. 5, September 1989, pp. 3593–3595.
2. Numerical computation of eddy currents in thin plates (A. Genon, J.P. Adriaens, P. Bourmanne, F. Delincé, W. Legros, **A. Nicolet**), *IEEE Transactions on Magnetics*, vol. 26, no. 5, September 1990, pp. 2376–2378.
3. Vector potential boundary element method for three dimensional magnetostatic (J.P. Adriaens, F. Delincé, P. Dular[†], A. Genon, W. Legros, **A. Nicolet**), *IEEE Transactions on Magnetics*, vol. 27, no. 5, September 1991, pp. 3808–3810.
4. Corner modelization for FEM-BEM Coupling (P. Bourmanne, F. Delincé, A. Genon, W. Legros, **A. Nicolet**), *IEEE Transactions on Magnetics*, vol. 27, no. 5, September 1991, pp. 4110–4113.
5. Skin effect and proximity effect in multiconductor systems with applied current and voltages (P. Bourmanne, F. Delincé, A. Genon, W. Legros, **A. Nicolet**), *Journal of Applied Physics*, 69(8), 15 April 1991, pp. 5035–5037.
6. Comparison of various methods for the modeling of thin magnetic plates (N. Bamps, F. Delincé, A. Genon, W. Legros, **A. Nicolet**), *Journal of Applied Physics*, 69(8), 15 April 1991, pp. 5047–5049.
7. Numerical computation of the magnetostriction effect in ferromagnetic materials (F. Delincé, A. Genon, J.M. Gillard, H. Hedia, W. Legros, **A. Nicolet**), *Journal of Applied Physics*, 69(8), 15 April 1991, pp. 5794–5796.
8. Graphical language interpreter and meshing tool for electromagnetic computations (N. Bamps, H. Hedia, A. Genon, W. Legros, **A. Nicolet**, M. Umé), *IEEE Transactions on Magnetics*, vol. 28, no. 2, March 1992, pp. 1771–1773.
9. Modelling of inductive effects in thin wire coils (F. Delincé, P. Dular[†], A. Genon, W. Legros, J. Mauhin, **A. Nicolet**), *IEEE Transactions on Magnetics*, vol. 28, no. 2, March 1992, pp. 1190–1192.
10. Line integrals for the computation of global parameters from 2D eddy current numerical computations (N. Bamps, A. Genon, W. Legros, H. Hedia, **A. Nicolet**), *IEEE Transactions on Magnetics*, vol. 27, no. 6, November 1991, pp. 4987–4991.
11. Couplage éléments finis - éléments frontières pour la modélisation du mouvement dans les structures bidimensionnelles (**A. Nicolet**, F. Delincé, A. Genon, W. Legros), *Journal de Physique III*, France 2 (November 1992) pp. 2035–2044.
12. Modelization of ferromagnetic materials in 2D finite element problems using Preisach's model (F. Henrotte, F. Delincé, **A. Nicolet**, A. Genon, W. Legros), *IEEE Transactions on Magnetics*, vol. 28, no. 5, September 1992, pp. 2614–2616.
13. Coupling between electric circuits and 2D magnetic field modelling (**A. Nicolet**, F. Delincé, N. Bamps, A. Genon, W. Legros), *IEEE Transactions on Magnetics*, vol. 29, no. 2, March 1993, pp. 1697–1700.
14. A coupling between edge finite elements, nodal finite elements and boundary elements for the calculation of 3-D eddy currents (P. Dular[†], A. Genon, W. Legros, J.-Y. Hody, J. Mauhin, **A. Nicolet**), *IEEE Transactions on Magnetics*, vol. 29, no. 2, March 1993, pp. 1470–1474.
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36. Comparison of various 2-D meshing techniques for finite element solutions (J.-F. Remacle, M. Umé, **A. Nicolet**, A. Genon, W. Legros). Presented at the International Workshop on Electric and Magnetic Fields, 18-20 May 1994, Leuven, published in *International Workshop on Electric and Magnetic Fields*, **A. Nicolet**, R. Belmans eds., Plenum Press, 1995.
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38. Magnetic field computation in devices including non linear material (**A. Nicolet**). Published in *Revue E of the Société Royale Belge des Electriciens*, 110th year, no. 4-94 (January 95), pp. 21–25.
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47. Integral methods for the characterisation of printed conductors (C. Millard, E. Sire, **A. Nicolet**), *EMF98*, Marseille, May 1998, conference proceedings, pp. 159–164.
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 57. Homogenization of non-Lax Milgram materials (F. Zolla, D. Felbacq, **A. Nicolet**, G. Bouchitté), PIERS 2004, Pisa, Italy, 2004.
 58. Diffraction par un cristal photonique métallique : dimension, polarisation et taille critique (F. Zolla, **A. Nicolet**), OHD'2003, Calais, 2003.
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 60. Analysis of optical metamaterials using the finite element method (**A. Nicolet**, F. Zolla), *Mathematics* 2005, Liverpool (UK), April 2005.
 61. Numerical analysis of optical metamaterials using the finite element method (**A. Nicolet**, F. Zolla), Acomen 2005, Ghent (Belgium).
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 64. Waveguide Propagation Modes and Quadratic Eigenvalue Problems (**A. Nicolet**, C. Geuzaine), CEM 2006, Aachen (Germany), April 2006.
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 66. A Scalar Finite Element Model for Spatial Solitons in Optical Fibres (**A. Nicolet**, F. Drouart, G. Renversez, C. Geuzaine), EMF 2006, May 2006, Aussois (France).
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 68. High order asymptotic analysis of twisted electrostatic problems (**A. Nicolet**, A.B. Movchan, C. Geuzaine, F. Zolla, S. Guenneau), Etopim 7, July 2006, Sydney (Australia).
 69. A finite element analysis of spatial solitons in optical fibres (F. Drouart, **A. Nicolet**, G. Renversez, C. Geuzaine), Etopim 7, July 2006, Sydney (Australia).
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 74. Finite element analysis of cylindrical invisibility cloaks of elliptical cross section (**A. Nicolet**, F. Zolla, S. Guenneau), Compumag 2007, June 2007, Aachen (Germany).
 75. Electromagnetic Analysis of Invisibility Cloaks (**A. Nicolet**, F. Zolla), Advanced Computational Electromagnetics workshop (ACE '07), June 2007, Aachen (Germany).
 76. How much can you twist a microstructured optical fibre? (**A. Nicolet**, F. Zolla, Y. Ould Agha), PIERS 2007, August 2007, Prague (République Tchèque).
 77. Geometrical Transformations and Equivalent Materials in Computational Electromagnetism (**A. Nicolet**, F. Zolla, Y. Ould Agha), ISEF 2007, September 2007, Prague (République Tchèque).
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 90. Pierre Godard, Frédéric Zolla and **André Nicolet**, Finite Element Modelling of Induced Gratings in nonlinear Optics, t. 8071, SPIE, 2011.
 91. Walasik, Wiktor, Kartashov, Yaroslav, **Nicolet, Andre** et al. 1D model for low power soliton hybridized with a plasmon in realistic nonlinear planar structures, 5th International Workshop on Theoretical and Computational Nano-Photonics (TaCoNa-Photonics 2012)Bad Honnef, GERMANY, October 24–26, 2012, AIP Conference Proceedings, volume: 1475, pp. 179–181, DOI: 10.1063/1.4750136, 2012.
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 94. Benjamin Vial, Guillaume Demésy, Frédéric Zolla, Mireille Commandré[†], **André Nicolet**, EMF 2013, Bruges (Belgium), April 2013, Finite element quasimodal analysis of Haroche's QED cavity.
 95. B. Vial, G. Demésy, F. Zolla, **A. Nicolet**, 3D FEM Quasimodal Analysis of Haroche QED Cavity, Compumag 2013, Budapest(Hungary), July 2013.
 96. F. Zolla, B. Vial, **A. Nicolet**, B. Guizal, D. Felbacq, Comment tordre la lumière à sa guise grâce à des métamatériaux, CNM 2013, Paris (France), October 2013 ;

97. B. Vial, **A. Nicolet**, F. Zolla, G. Demésy, Exact PML and the numerical computation of quasi-modes in electromagnetic open structures, URSI Radio 2014, Mauritius, April 2014, book of abstracts p. 50.
98. Guillaume Demésy, **André Nicolet**, Frédéric Zolla, Benjamin Vial, Mohamed Hamed, Sulaimon Balogun, Quasimodal analysis of periodic structures and discontinuous waveguides, Book of abstracts of the 6th International Conference on Advanced Computational Methods in Engineering, ACOMEN 2014, Ghent (Belgium), June 2014.
99. **A. Nicolet**, D. Demésy, F. Zolla, B. Vial, Quasimodal analysis of segmented waveguides, 2014 IEEE International Conference on Antenna Measurements & Applications (CAMA), Juan-les-Pins (France), 6-19 Nov. 2014, pp. 1–4, INSPEC Accession Number: 14841924
100. Quasi-modes in segmented waveguides, D. Demésy, **A. Nicolet**, F. Zolla, PIERS 2015, Prague (Czech Republic), July 6-9, 2015.
101. A.-S. Bonnet-Ben Dhia, C. Carvalho, C. Chambeyron, L. Chesnel, P. Ciarlet, **A. Nicolet**, F. Zolla, Curious energy losses at corners of metallic inclusions, Waves 2015, Karlsruhe (Germany), July 2015.
102. Quasimode computation in structures including several dispersive materials (G. Demésy, M. Garcia-Vergara, F. Zolla, **A. Nicolet**) Conference on Metamaterials XI, Prague, Czech Republic, April 26–27, 2017, Proceedings of SPIE, volume: 10227, Article Number: UNSP 102270G, 2017.

4.4.5 Invited seminars and lectures

1. *Finite Element Modelling of Twisted Microstructured Optical Fibres*, Applied Mathematics Seminars, University of Liverpool, March 2005;
2. *A Covariant Formulation of Electromagnetism with an Application to the Modelling of Twisted Waveguides*, Applied Mathematics Seminars, University of Liverpool, October 2005;
3. *Introductory Lecture: FE Modelling of Wave Guides*, CEM 2006, Aachen (Germany), April 2006;
4. *Twisted Waveguides and Geometrical Transformations*, Advanced Computational Electromagnetics workshop (ACE'06), Boston, May 2006;
5. *Invisibility Cloaks of Arbitrary Shapes*, Applied Mathematics Seminars, University of Liverpool, December 2006;
6. *Geometrical Transformations for Numerical Modelling and for New Material Design in Photonics*, Meta'08, Marrakech, May 2008;
7. *Geometrical transformations in photonics*, International Workshop on "Hamiltonian Approaches of ITER Physics", CIRM (Centre International de Rencontres Mathématiques), Marseille, France, 2–6 November 2009;
8. *Masking with Generalized Cloaking*, Compumag 2009, Florianópolis, Brazil, 22–26 November 2009;
9. *Invisibility cloaks and transformation optics*, LPNHE seminar, Paris, 4 March 2010;
10. *Invisibility cloaks and transformation optics*, INRIA Sophia Antipolis seminar, April 2011;
11. *Invisibility cloaks and transformation optics*, seminar at Centre Spatial de Liège (CSL), University of Liège, Belgium, March 2012;
12. Invited lecturer at "GeMs&ACEs 2014: Short Course – Geometric Methods and Applied Computational Electromagnetics", six-hour lecture *Numerical Modelling in Optics*, A Coruña (Spain), June 2014;
13. *Quasi-modal analysis of segmented waveguides*, 2014 IEEE International Conference on Antenna Measurements & Applications, Juan-les-Pins (France), November 2014;
14. *Resonances in open electromagnetic devices: (simple) theory, numerical computation and applications*, seminar at Laboratoire Photonique Numérique et Nanosciences, Institut d'Optique, Bordeaux, March 2015;
15. *Resonances in open electromagnetic devices: (simple) theory, numerical computation and applications*, ENSTA ParisTech (Palaiseau), Leaky Days, 25–26 June 2015;
16. *Transformation optics I & II*, lectures at the 2nd Workshop on Seismic Metamaterials, Dakar (Senegal), May 2016;
17. *Quasi-normal modes and their applications*, talk at the 2nd Workshop on Seismic Metamaterials, Dakar (Senegal), May 2016;
18. *Finite Elements for Electrodynamics and Modal Analysis of Dispersive Structures*, Mathematical and Numerical Modeling in Optics, Workshop of the IMA (Institute for Mathematics and its Applications), Minneapolis (USA), December 2016;
19. *Topology for Classical Electrodynamics: Maxwell equations, numerical modeling, and topological photonics* (parts 1, 2 and 3), three lectures at the Topical School 2017 "Curved Spacetimes analogs in Condensed Matter Physics", Nancy, Faculté des Sciences, 4–6 October 2017, Doctoral College "Statistical Physics of Complex Systems", L4 Collaboration Coventry – Leipzig – Lviv – Nancy;
20. *Exact Dispersive Quasi-normal Mode (DQNM) Expansion of Dispersive Nanophotonics Structures*, INRIA Sophia Antipolis, March 2020;
21. **Workshop Imperial College – Institut Fresnel on Plasmonics & Metamaterials, 27–28 September 2021: From Finite Elements to Dispersive Quasi-normal Modes (DQNM) in Nanophotonics, 28 September 2021.**
At this first joint workshop, I gave the Institut Fresnel keynote, the Imperial College keynote being given by Sir John Pendry.
22. *Quasi-normal Mode Expansion in Unbounded Photonic Structures: Perfectly Matched Layers (PML) vs. Bayliss–Turkel (BT) Absorbing Boundary Conditions*, META 2023, Paris, July 2023;
23. **In September and October 2024, at the request of several colleagues, I gave four seminars (eight hours in total) at Institut Fresnel on differential geometry and exterior calculus applied to electro-**

magnetism and its numerical modelling.

4.4.6 Outreach and science communication

In 2007, my research on the “invisibility cloak” received wide coverage in the general and popular-science press. I was interviewed, for instance, by:

- Philipp Bornebusch, Wissenschaft Online GmbH / spektrumdirekt.de;
- Cyrus Farivar, wired.com;
- Frédéric Garlan, Agence France-Presse (AFP);
- Gaëlle Le Garrec, “Futurs” pages of *Le Monde*;
- Jérôme Fénoglio, *Le Monde* (a full-page article with a summary on the front page, 18 May 2007);
- Lionel Dian, *Marseille l'Hebdo*;
- Caroline Lachowsky, Radio France Internationale (RFI);
- David Larousserie, *Sciences & Avenir* (cover and special feature in the October 2007 issue);
- Fabien Gruhier, *Le Nouvel Observateur*;
- the LCM television news;
- Sophie Manelli, daily newspaper *La Provence*;
- Laurence Mildonian, *Marseille l'Hebdo*;
- Lætitia Theunis, *Le Soir* (Brussels, Belgium), article of 25 August 2014;

which led to several articles and radio or television programmes. I am also regularly consulted as an expert by the magazine *La Recherche* (Hélène Le Meur, deputy editor-in-chief, and her colleagues) and the online magazine Futura-Sciences.com (Laurent Sacco).

I published an article for a broad audience of physicists in the CNRS review *Images de la physique 2010*: “*Les capes d’invisibilité et l’optique transformationnelle*” (invisibility cloaks and transformation optics), A. Nicolet.

I have also advised several groups of high-school and preparatory-class students on their TIPE projects on invisibility, and I took part as a speaker in the *Fête de la Science*.

Public lecture (1.5 h) “L’invisibilité, mirage théorique ou vision du futur ?” (Invisibility: theoretical mirage or vision of the future?) at the Université du Temps Libre, Aix-en-Provence, 21 March 2019.

5 ADMINISTRATION AND OTHER COLLECTIVE RESPONSIBILITIES

5.1 Teaching and administration

- Head of the ATHENA team of Institut Fresnel from its creation in March 2015, for a four-year term.
- Co-leader (with B. Gralak), 2015–2018, of the research theme “Electromagnetism and Metamaterials” (one of the four themes) of Institut Fresnel.
- Director of studies of the DESS (professional postgraduate degree) in Electrical Engineering and Industrial Computing from its creation in September 2001 to September 2005; I created this degree with Prof. A. Mouillet in September 2001.
- Member of the advisory board (*Conseil de Perfectionnement*) of the IUP GEII of Aix-Marseille III from its creation in 2000 until the degree was discontinued in 2006.
- Member of the teaching team of the Bachelor in Physical and Chemical Sciences, Université d’Aix-Marseille III.
- Aix-Marseille III correspondent and member of the board of the Research Master in Theoretical Physics, Mathematical Physics and Particle Physics of Aix-Marseille from 2003.
- Elected member of the Laboratory Council of Institut Fresnel for a four-year term from June 2006.
- Member of the recruitment committee (*commission de spécialistes*) of the 63rd section, Université Paul Cézanne Aix-Marseille III, 1997–2011.
- External member of the recruitment committee of the 61st/63rd sections, Université de Provence Aix-Marseille I, 1997–2011.
- External member of the recruitment committee of the 63rd section, Université de Toulon, 2007.
- Vice-chair of the Physics Committee, June 2000 – May 2002.

5.2 Scientific and editorial activities

5.2.1 PhD thesis committees

- Djillali Benzerga (Université Paris 6, 1993) *referee*;
Contribution à la modélisation 3D des phénomènes électromagnétiques régis par les équations de la magnétostatique et de la magnétodynamique
- Olivier Ninet (Université Claude Bernard, Lyon I, 1996) *referee*;
Prise en compte du phénomène d'hystérésis dans un logiciel de calcul de champ 2D en magnétostatique : validation expérimentale
- Dominique Leschi (Ecole Centrale de Lyon, 1996) *referee*;
Modélisation et optimisation des inducteurs dans les systèmes de cuisson par induction
- Fabienne Cortial (INP Grenoble, 1996) *referee*;
Modélisation de l'hystérésis et des dispositifs d'enregistrement magnétique
- Jean-François Remacle (University of Liège, Belgium, 1997);
Estimation d'erreur dans la modélisation par éléments finis des champs électromagnétiques : Application à l'optimisation des maillages
- Cristian Golovanov (INP Grenoble, 1997) *referee*;
Développement de formulations éléments finis 3D en potentiel vecteur magnétique : application à la simulation de dispositifs électromagnétiques en mouvement
- Cathy Barnetche (INP Lorraine, 1997) *referee*;
Calcul numérique de machines asynchrones en régime transitoire, en tenant compte de la saturation magnétique
- Stefan Henneberger (KU Leuven, Belgium, 1998);
Design and Development of a Permanent Magnet Synchronous Motor for a Hybrid Electric Vehicle Drive
- Moez Youssef (INP Grenoble, 1998) *referee*;
Rayonnement dans les convertisseurs d'électronique de puissance
- Yvonnick Le Menach (U. Lille, 1999) *referee*;
Contribution à la modélisation numérique tridimensionnelle des systèmes électrotechniques : prise en compte des inducteurs
- Fabiano Luiz Santos Garcia (INP Grenoble, 1999) *referee*;
Modélisation des composants électromagnétiques haute fréquence par la méthode des éléments finis
- Benoît Meys (University of Liège, Belgium, 1999);
Modélisation des champs électromagnétiques aux hyperfréquences par la méthode des éléments finis : application aux problèmes de chauffage diélectrique
- Céline Brasseur Millard (Université d'Aix-Marseille III, 1999);
Modélisation des capacités parasites des éléments bobinés
- Fouad Azzouz (Université de Nantes, 2000) *referee*;
Contribution à la modélisation de tôles ferromagnétiques par impédance de surface et éléments finis : application au calcul des puissances du chauffage par induction
- Yacine Azzouz (Université d'Aix-Marseille III, 2000);
Nouvelle méthode de caractérisation des tôles magnétiques en champ tournant
- Sébastien Guenneau (Université d'Aix-Marseille III, 2001);
Homogénéisation des quasi-cristaux et analyse des modes dans des fibres optiques de type cristal photonique
- Alita Dewi (INP Grenoble, 2001) *referee, chair*;
Apport des nouvelles technologies interactives pour l'analyse intégrée en génie électrique : vers un laboratoire virtuel d'expérimentation en électrotechnique
- Corinne Duloung (Université d'Aix-Marseille III, 2001);
Modélisation comportementale des perturbations électromagnétiques en électronique de puissance : signature CEM d'une cellule de commutation

- Olivier Barré (Ecole Centrale de Lille and Université des Sciences et Techniques de Lille, 2003) *referee*;
Contribution à l'étude des formulations de calcul de la force magnétique en magnétostatique, approche numérique et validation expérimentale
- Thomas Hamon (Université de Versailles, 2007) *referee*;
Cristaux magnéto-photoniques unidimensionnels
- Abdechafik Hadjadj (Université d'Aix-Marseille III, 2007) *chair*;
Contribution au développement d'une méthode de détection de défauts par approche photothermique dans les liaisons filaires blindées
- Yacoub Ould Agha (Université d'Aix-Marseille I, 2007) *chair*;
Transformations géométriques réelles et complexes : application à la recherche de modes à pertes dans des fibres optiques microstructurées
- Fabien Drouart (Université d'Aix-Marseille III, 2008);
Étude de la non-linéarité Kerr dans les fibres optiques microstructurées
- Anass Aïssi (Université d'Aix-Marseille I, 2008) *chair*;
La modélisation des tubes à onde progressive à hélice en domaine temporel
- Yannick Vuillermet (INP Grenoble, 2008) *chair*;
Immunsation en boucle fermée — application au sous-marin double coque : identification de l'aimantation à partir de mesures du champ magnétique proche
- Guillaume Demésey (Université d'Aix-Marseille III, 2009);
Modélisation électromagnétique tri-dimensionnelle de réseaux complexes. Application au filtrage spectral dans les imageurs CMOS
- Cédric Blanchard (University of Granada, 2009);
Simulation of electromagnetic waves propagating in complex media with the TLM method
- Pierre Godard (Université d'Aix-Marseille I, 2009);
Optique électromagnétique non-linéaire polyharmonique : théorie et modélisation numérique
- M. Malidé (Université d'Aix-Marseille III, 2010);
Modélisation électromagnétique tridimensionnelle
- Hicham Belyamoun (Supélec, Paris, 2010) *referee*;
Modélisation et caractérisation spectrale de métamatériaux
- Benjamin Goursaud (ENSTA, Paris, 2010);
Étude mathématique et numérique de guides d'ondes ouverts non uniformes, par approche modale
- Christophe Bourel (Université du Sud Toulon-Var, 2010) *referee*;
Étude mathématique et numérique de cristaux photoniques fortement contrastés
- Guillaume Wasselynck (Université de Nantes, 2011) *referee*;
Étude des interactions entre les ondes électromagnétiques de fréquences moyennes et les matériaux composites : application à l'assemblage par induction de ces matériaux
- Pierre Bernardi (Université d'Aix-Marseille I, 2011);
Utilisation et Amélioration du Modèle Discret d'Excitation d'un Guide d'Onde Périodique pour la Simulation Pratique du Tube à Onde Progressive en Domaine Temporel
- Clément Durochat (INRIA, Sophia-Antipolis, 2013);
Méthode de type Galerkin discontinu en maillages multi-éléments (et non-conformes) pour la résolution numérique des équations de Maxwell instationnaires
- Mandiaye Fall (AMU - CEA Leti, Marseille, 2013) *chair*;
Modélisation multi-échelle de systèmes nanophotoniques et plasmoniques
- Yvan Voznyuk (AMU, Marseille, 2014);
Méthodes de décomposition de domaines pour la modélisation électromagnétique et l'imagerie micro-onde quantitative dans des configurations tridimensionnelles de grande taille

- François Caire (Université Paris Sud 11 - CEA LIST/DISC Saclay, 2014) *referee*;
Les équations de Maxwell covariantes pour le calcul rapide des champs diffractés par des conducteurs complexes. Application au contrôle non destructif par courants de Foucault
- Raphaël Scapolan (Université Grenoble Alpes, Grenoble, 2014) *referee*;
Modélisation électromagnétique 3D d'inducteurs multibrins
- Khac Long Nguyen (IFSTTAR - Centre de Nantes, 2014) *referee*;
Modélisation numérique de la propagation des ondes élastiques guidées dans des structures courbes ouvertes et enfouies
- Vincent Debierre (Ecole Centrale de Marseille, Marseille, 2015);
La fonction d'onde du photon en principe et en pratique
- Changyou Li (Université Paris Sud, Paris, 2015) *referee*;
Modélisation et simulation de la diffraction électromagnétique par des laminés plans renforcés par des fibres cylindriques arrangées périodiquement
- Nicolas Marsic (Université de Liège, Liège, Belgium, 2016);
Efficient methods for large-scale time-harmonic wave simulations
- Stéphane Théveny (AMU, Marseille, 2016) *chair*;
Approches fréquentielle et temporelle de la dynamique des tubes à onde progressive
- Zicheng Liu (Université Paris-Saclay, 2017) *chair*;
Electromagnetic modeling and imaging of damages of fiber-reinforced composite laminates
- Damien Minenna (Aix-Marseille Université, 2019);
Modélisation hamiltonienne N-corps de l'échange de moment dans l'interaction onde-particule non-linéaire
- Houssem Chebbi (CEA Saclay, 2020) *referee*;
Méthode des coordonnées curvilignes pour la modélisation électromagnétique des matériaux complexes : application au contrôle non destructif par courants de Foucault des matériaux composites
- Minh Duy Truong (Aix-Marseille Université, 2020);
Théorie et modélisation numérique des résonances photoniques : expansion modale quasi-normale : applications en théorie électromagnétique
- Alexandre Gras (Université de Bordeaux, 2021) *chair*;
Reconstruction des champs électromagnétiques avec les modes quasinormaux : une approche numérique
- Mariano Pascale (University of Naples "Federico II", 2021) *evaluation report*;
Material-Independent Modes for the Electromagnetic Scattering From Homogeneous Objects
- Jonathan Velasco (Université de Liège, 2022);
Finite-Element Modeling of Thin Wires Including Skin- and Proximity Effects
- Igor Glukhov (ENIB, Brest, 2022) *referee*;
Manipulation of electromagnetic waves in the visible, IR and microwave domains using plasmonic heterostructures and beam properties
- Massimiliano Montone (INRIA Sophia Antipolis, Université de Nice, 2023).
Méthodes Galerkin discontinues en domaine temporel pour la simulation full-wave de dispositifs à semi-conducteur

5.2.2 Habilitation (HDR) committees

- Yves du Terrail Couvat (INP Grenoble, 2001) *referee*;
Modélisation numérique multiphysique appliquée aux procédés de fusion en creuset froid inductif
- Frédéric Zolla (Université d'Aix-Marseille I, 2004);
Contribution à l'étude de la diffraction et de la propagation des ondes électromagnétiques : Homogénéisation des structures périodiques et Propagation dans des fibres microstructurées
- Éric Akmansoy (Université Paris-Sud, 2010) *referee*;
Cristaux photoniques et métamatériaux en vue de leurs applications
- Salim Boutami (CEA Grenoble, 2018) *referee*;

- Fabien Treyssède (Le Mans Université, 2019) *referee*;
Contributions à la modélisation numérique des guides d'ondes élastiques
- Anastassios Skarlatos (CEA Saclay, 2023) *referee*;
Near-field scattering solutions using mixed bases with application to electromagnetic and thermographic inspection of technical materials
- Cédric Blanchard (Université d'Orléans, 2024) *referee*.
Contrôle des propriétés thermo-optiques d'ensembles désordonnés de nanoparticules en interaction résonante : vers l'échelle de la longueur d'onde

5.2.3 Reviewer for the following journals

- Science;
- Optics Letters;
- Optics Express;
- JOSA A;
- IEEE Microwave Theory and Techniques;
- IEEE Transaction on Magnetics;
- IEEE Transactions on CAD of Integrated Circuits and Systems (TCAD);
- IEEE Microwave and Wireless Components Letters;
- Proceedings of the Royal Society A ;
- Waves in Random and Complex Media;
- COMPEL;
- Journal of Electromagnetic Waves and Applications;
- Journal de Physique III;
- Physica B;
- Quarterly Jnl. of Mechanics & App. Maths;
- Applied Physics Letters;
- Journal of Chinese Physics Letters;
- The European Physical Journal Applied Physics (EPJAP);
- Journal of Computational Physics;
- Advances in Computational Mathematics;
- Transport Theory and Statistical Physics;
- International Journal for Numerical Methods in Engineering;
- Numerical Methods for Partial Differential Equations;
- International Journal of Numerical Modelling: Electronic Networks, Devices and Fields;
- Journal of Applied Geophysics;
- Journal of Modern Optics;
- Journal of Nanophotonics;
- International Journal for Applied Electromagnetics and Mechanics;
- Radio Science;
- Discrete Dynamics in Nature and Society;
- European Journal of Physics;
- Journal of Engineering Research;
- Mathematics and Computers in Simulation;
- Computer Methods in Applied Mechanics and Engineering;
- European Journal of Mechanics / A Solids;
- New Journal of Physics;
- Journal of Lightwave Technology;
- Photonics Journal;
- Journal of Computational Electronics;
- Journal of Magnetism and Magnetic Materials.

5.2.4 Conference organisation and scientific committees

- **International Workshop on Electric and Magnetic Fields (EMF):**

I organised this conference, planned in Marseille, fully online (because of COVID-19) in July 2020, with C. Geuzaine.

I then organised the conference in Marseille in August 2023. Member of the Organising Committee and of the Scientific Committee since its creation in 1992: Liège, 28–30 September 1992 (**initiator and organiser of this first edition**); Leuven, 17–20 May 1994, and co-editor of the proceedings, *Electric and Magnetic Fields: From Numerical Models to Industrial Applications*, A. Nicolet, R. Belmans (eds.), Plenum Press, New York, 1995; Liège, 7–10 May 1996; Marseille, France, 12–15 May 1998 (which I organised and co-chaired with P. Dular[†] of the University of Liège); Ghent, 17–19 May 2000; Aachen, September 2003; Aussois, June 2006; Mondovì, May 2009; Bruges, April 2013.

- **COMPUMAG conferences:**

For Compumag 2019 (Paris), 2021 (Cancún), 2023 (Kyoto) and 2025 (Naples), Track Editor in charge of about thirty submissions, from the selection of reviewers to the final publication of the accepted papers. Member of the Editorial Board since 1995 (IEEE publications); session chair at Compumag 1993 (Miami), 1995 (Berlin), 2001 (Évian), 2007 (Aachen), 2009 (Florianópolis), 2013 (Budapest) and 2017 (Daejeon).

- **Numélec conference:**

Member of the Scientific Committee since 2007 (still serving in 2026), vice-chair (2008–2012) and chair (2012–2015) of the Scientific Committee. Organiser of Numélec 2012 in Marseille (chair of the Local Organising Committee).

- Member of the scientific committee of the annual TaCoNa-Photonics conference 2012 (Physikzentrum, Bad Honnef, Germany).
- Member of the scientific committee of the French National Metamaterials Conference (CNM) 2011 (March 2011), CNM 2013 (October 2013) and CNM 2017 (March 2017), Paris, France.
- Member of the Program Committee of PHOTOPTICS 2014, 2015, 2016 and 2017.
- Symposium on Electromagnetic Phenomena in Nonlinear Circuits (EPNC): member of the International Organising Committee since 1994; Poznań, Poland, June 1996; Liège, Belgium, September 1998; Leuven, Belgium, July 2002; session chair at Poznań, Poland, June 2004, and at Maribor, June 2006.
- Member of the Scientific and Editorial Committee of ISEM 2003, Versailles, France, October 2003, and session chair.
- CEFC conferences: member of the Editorial Board since 1994 (IEEE publications); session chair at CEFC 1994, Aix-les-Bains.
- Co-chair of the conference Betech 95, Liège, 13–15 September 1995, co-editor of the proceedings (*Boundary Element Technology X*, Computational Mechanics Publications) and session chair.
- Member of the Scientific Committee of the Colloquium *Méthodes informatiques de la conception industrielle*, Marseille, France, 18 June 1993.
- Organiser of the CREMIS day (Centre de Rencontres et d'Échanges en Modélisation pour l'Ingénierie Scientifique), 15 June 1995, on *Electromagnetic phenomena studied by coupling physical models*.
- Secretary of the FNRS Contact Group “Boundary Element Method” from 1991 to 1993, which fostered exchanges between the Belgian laboratories using the boundary element method through meeting days.

5.2.5 Expert reviewing (France)

- ANR: member of the Scientific Evaluation Committee “Mathematics, Computer Science, Systems and Communication Engineering” (CES 40), 2015.
- Expert reviewer for ANR (2010, 2011, 2012, 2013, 2014, 2015).

5.2.6 Expert reviewing (international)

- Expert reviewer for the Marsden Fund, New Zealand (since 2011);
- Expert reviewer for City University of Hong Kong, China (2011);
- Expert reviewer for Tampere University of Technology, Finland (2015);

- Expert reviewer for the FWF – Austrian Science Fund (2018, 2023);
- Member of the Scientific Advisory Board of the EPSRC Programme on Metamaterials (Department of Mathematics, Imperial College London (Prof. R. Craster); Department of Physics, Imperial College London (Prof. S. Maier, Sir J. Pendry); Department of Mathematical Sciences, University of Liverpool (Prof. A. Movchan)), **taking part in the six annual Board meetings in London from 2014 to 2019.**

5.2.7 Pedagogy

- AMU-CIPE training “Teaching in a flipped classroom” (March 2023), put into practice in the course *Formal Tools for Physics* (L1, semester 2).
- Creation, coordination and teaching in the international multidisciplinary Master in Modelling, created jointly by the Faculty of Sciences and Techniques of UCAD and the Faculty of Sciences of Aix-Marseille Université (AMU). The cohesion of the teaching team is maintained through frequent online meetings and visits in both directions. From 18 to 24 May 2025, I stayed at UCAD with other AMU members of the teaching team, funded by an Erasmus+ teaching mobility grant (STA).